Analysis of Factors Associated with Poor Response to Nutritional Treatment in Severely Malnourished Children Aged 6–59 Months Without Edema and With Complications Treated (Phase T).

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1. Methodology

Descriptive analyses were conducted to characterize the cohort and the non-response to the nutritional treatment. Continuous variables were summarized using medians and interquartile ranges (25th - 75th percentile) due to their non-normal distribution, whereas categorical variables were summarized as counts and percentages (%).

Differences between groups (success and failure) were assessed using the Wilcoxon rank-sum test or Kruskal–Wallis test for continuous variables and Pearson's chi-square or Fisher's exact test for categorical variables, as appropriate.

Univariable logistic regression analyses were performed to estimate crude associations between independent variables and the dependent variable. Variables considered clinically relevant or statistically associated with the non-response to the nutritional treatment at $\underline{p} < 0.20$ were included in a multivariable logistic regression.

Prior to model building, pairwise correlations among independent variables were examined to identify highly correlated predictors and assess potential multicollinearity. Pearson correlation coefficients were computed for continuous variables (or Spearman's rank correlations for non-normally distributed variables). Variables exhibiting strong correlations ($|r| \geq 0.70$) were carefully evaluated to avoid redundancy. Variance Inflation Factors (VIFs) were subsequently calculated in the multivariable model, with VIF values > 5 indicating potential multicolliniarity.

Model selection was guided by likelihood ratio tests and the principle of parsimony. A stepwise selection procedure was used to identify the most parsimonious model (based on the Akaike Information Criterion (AIC)) by retaining variables that improved model fit while excluding predictors that contributed little additional information.

Adjusted odds ratios (aORs) with 95% confidence intervals (CIs) were reported. Statistical significance in the final model was defined as a two-sided $\underline{p} < 0.05$. A variable was significantly associated with the risk of nutritional treatment failure if: i) the odds ratio (OR) was > 1 with a lower limit of the confidence interval (CI) > 1; or ii) OR<1 with upper CI limit <1; and p-value was <0.05.

All analyses were conducted using R (version 4.3.0).

2. Study population

Insert comments here.

3. Descriptive Characteristics

Insert comments here.

4. Bivariate Analysis

Table 1: Association between explanatory variables and outcome at phase T.

	Non-response to treatment in Phase				
Characteristic	**FALSE** N = 634[1]	**TRUE** N = 126[1]	**Overall** N = 760[1]	**p-value**[23]	**q-value**[4]
Age (Month)				0.128	0.207
Median (Q1 ; Q3)	14 (10.0 ; 22)	17 (11.0 ; 22)	15 (11.0 ; 22)		
Min ; Max	6 ; 59.0	7 ; 48.0	6 ; 59.0		
Age group				0.942	0.971
[25;59]	114 (18.0%)	23 (18.3%)	137 (18.0%)		
[6;25[	520 (82.0%)	103 (81.7%)	623 (82.0%)		
Gender				0.134	0.207
Male	363 (57.3%)	63 (50.0%)	426 (56.1%)		
Female	271 (42.7%)	63 (50.0%)	334 (43.9%)		
Locality				0.880	0.965
Urban	147 (23.2%)	30 (23.8%)	177 (23.3%)		
Rural	487 (76.8%)	96 (76.2%)	583 (76.7%)		
Vaccination status				**0.045***	0.089
Up to date	217 (80.1%)	38 (67.9%)	255 (78.0%)		
Out of date	54 (19.9%)	18 (32.1%)	72 (22.0%)		
Missing values	363	70	433		
Breastfeeding status	529 (95.3%)	106 (95.5%)	635 (95.3%)	0.934	0.971
Missing values	79	15	94		
Type admission				0.626	0.760
New admission	543 (85.6%)	110 (87.3%)	653 (85.9%)		
Readmission	91 (14.4%)	16 (12.7%)	107 (14.1%)		
Morbidity burden				**0.033***	0.080
Median (Q1 ; Q3)	2 (1.0 ; 3)	3 (2.0 ; 3)	2 (1.0 ; 3)		
Min ; Max	0 ; 6.0	0 ; 6.0	0 ; 6.0		

Characteristic	Non-response to treatment in Phase				
	FALSE N = 634[1]	**TRUE** N = 126[1]	**Overall** N = 760[1]	**p-value**[23]	**q-value**[4]
Shock	13 (2.1%)	4 (3.2%)	17 (2.2%)	0.504	0.635
Gastroenteritis	429 (67.7%)	101 (80.2%)	530 (69.7%)	**0.005****	0.021
Pneumonia	375 (59.1%)	87 (69.0%)	462 (60.8%)	**0.038***	0.085
Tuberculosis	4 (0.6%)	1 (0.8%)	5 (0.7%)	>0.999	>0.999
Digestive candidiasis	90 (14.2%)	26 (20.6%)	116 (15.3%)	0.066	0.119
HIV	10 (1.6%)	4 (3.2%)	14 (1.8%)	0.267	0.363
Severe malaria	299 (47.2%)	45 (35.7%)	344 (45.3%)	**0.018***	0.048
Neurological malaria	30 (4.7%)	1 (0.8%)	31 (4.1%)	**0.041***	0.088
Severe anemia	161 (25.4%)	47 (37.3%)	208 (27.4%)	**0.006****	0.021
Other anemia	161 (25.4%)	47 (37.3%)	208 (27.4%)	**0.006****	0.021
Incident edema (PT)	0 (0.0%)	3 (2.4%)	3 (0.4%)	**0.004****	0.021
Phase length (PT)				0.720	0.844
Median (Q1 ; Q3)	2 (1.0 ; 3)	2 (1.0 ; 3)	2 (1.0 ; 3)		
Min ; Max	1 ; 11.0	1 ; 8.0	1 ; 11.0		
Weight gain g/kg/day (PT)				**<0.001*****	<0.001
Median (Q1 ; Q3)	12 (7.5 ; 15)	5 (3.1 ; 8)	9 (5.3 ; 13)		
Min ; Max	5 ; 36.0	1 ; 16.6	1 ; 36.0		
Missing values	583	100	683		
Phase T to Phase 1 transition	0 (0.0%)	102 (81.0%)	102 (13.4%)	**<0.001*****	<0.001
Nutritional treatment (PT)				0.201	0.297
F75+RUTF/F75F+RUTF	347 (54.7%)	57 (45.2%)	404 (53.2%)		
F100	92 (14.5%)	22 (17.5%)	114 (15.0%)		
F75/F75F	58 (9.1%)	17 (13.5%)	75 (9.9%)		
Others	137 (21.6%)	30 (23.8%)	167 (22.0%)		
F75+RUTF to F100 transition (PT)	0 (0.0%)	5 (4.0%)	5 (0.7%)	**<0.001*****	<0.001
F75 frequency below target (PT)				0.441	0.576
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 0.0	0 ; 1.0		
F75 frequency above target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F75 quantity below target (PT)				0.835	0.946
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		

Characteristic	Non-response to treatment in Phase				
	FALSE N = 634[1]	**TRUE** N = 126[1]	**Overall** N = 760[1]	**p-value**[2,3]	**q-value**[4]
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 quantity above target (PT)				0.225	0.319
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence below target (PT)				**0.016***	0.047
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F75 adherence above target (PT)				0.123	0.207
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 frequency below target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100 frequency above target (PT)				**0.011***	0.032
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 1)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 quantity below target (PT)				0.056	0.105
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 quantity above target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
F100 adherence below target (PT)				**<0.001*** **	0.004
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 1.0	0 ; 1.0	0 ; 1.0		
F100 adherence above target (PT)					
Median (Q1 ; Q3)	0 (0.0 ; 0)	0 (0.0 ; 0)	0 (0.0 ; 0)		
Min ; Max	0 ; 0.0	0 ; 0.0	0 ; 0.0		
Death (PT)	0 (0.0%)	3 (16.7%)	3 (0.6%)	**<0.001*** **	<0.001
Missing values	128	108	236		
Type of discharge (PT)				**<0.001*** **	<0.001
Recovered/Transferred	485 (95.8%)	13 (72.2%)	498 (95.0%)		
Absconded	19 (3.8%)	2 (11.1%)	21 (4.0%)		

Characteristic	Non-response to treatment in Phase				
	FALSE N = 634[1]	**TRUE** N = 126[1]	**Overall** N = 760[1]	**p-value**[23]	**q-value**[4]
Death	0 (0.0%)	3 (16.7%)	3 (0.6%)		
Medical referral	2 (0.4%)	0 (0.0%)	2 (0.4%)		
Missing values	128	108	236		

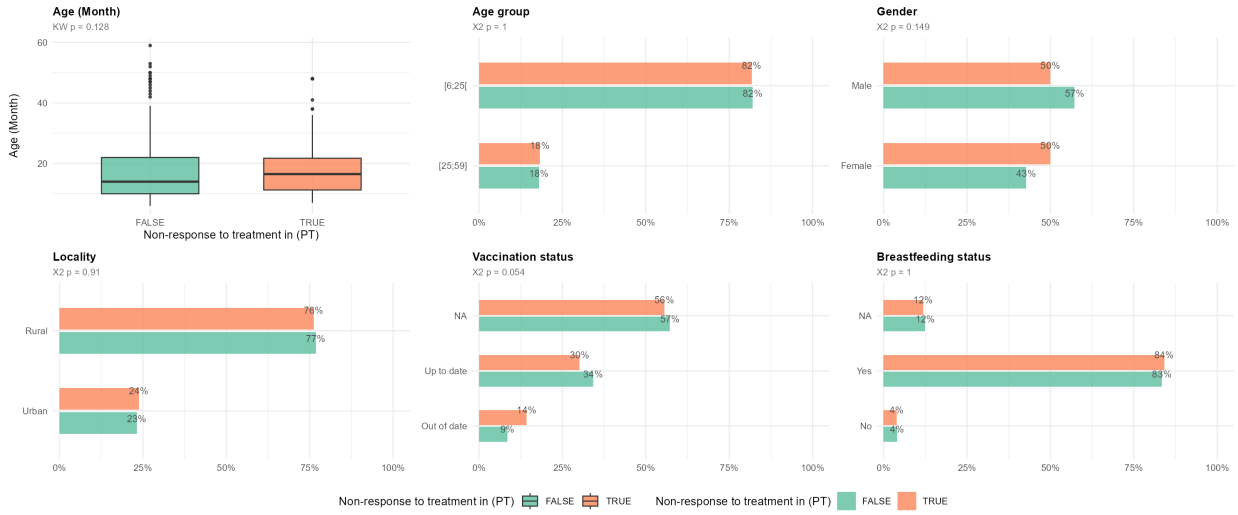
[1] n (%)

[2] Wilcoxon rank sum test; Pearson's Chi-squared test; Fisher's exact test; NA

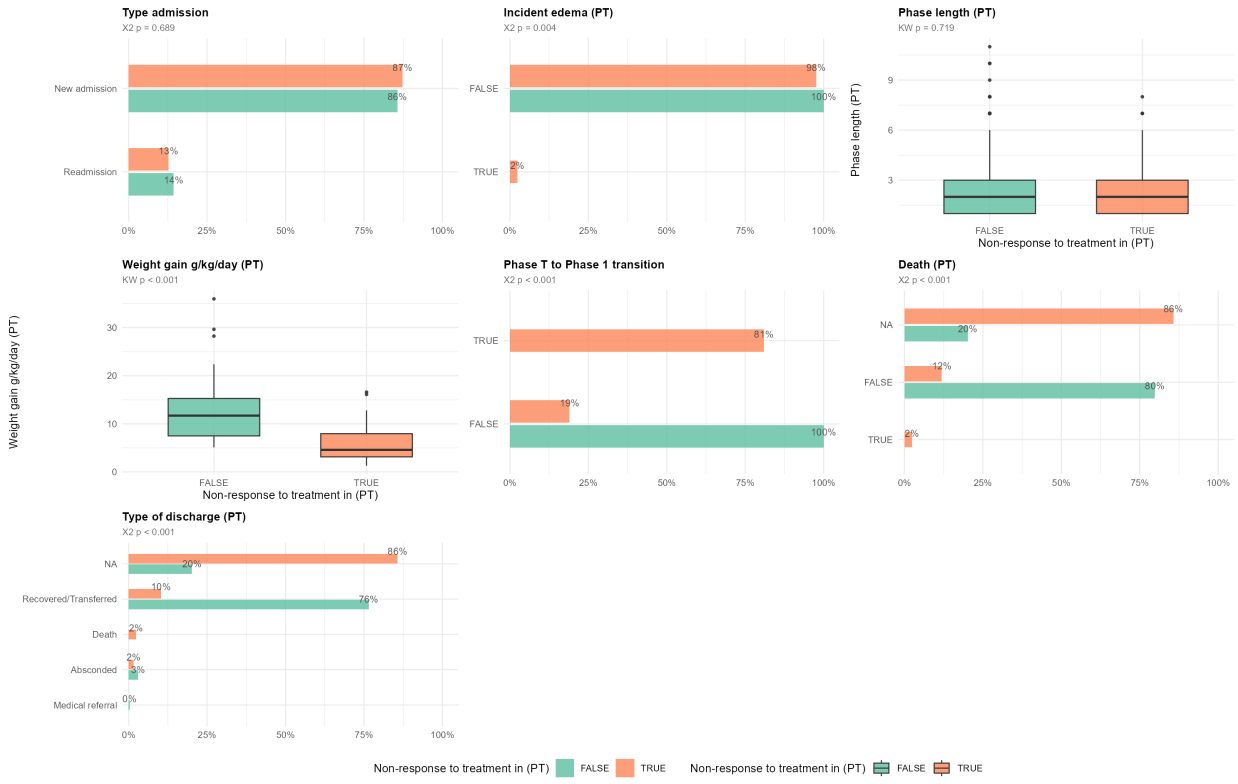
[3] *p<0.05; **p<0.01; ***p<0.001

[4] False discovery rate correction for multiple testing

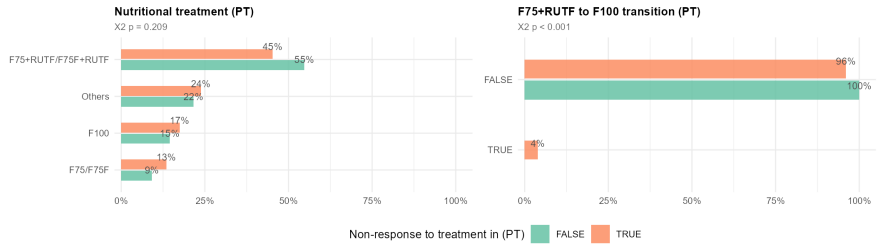
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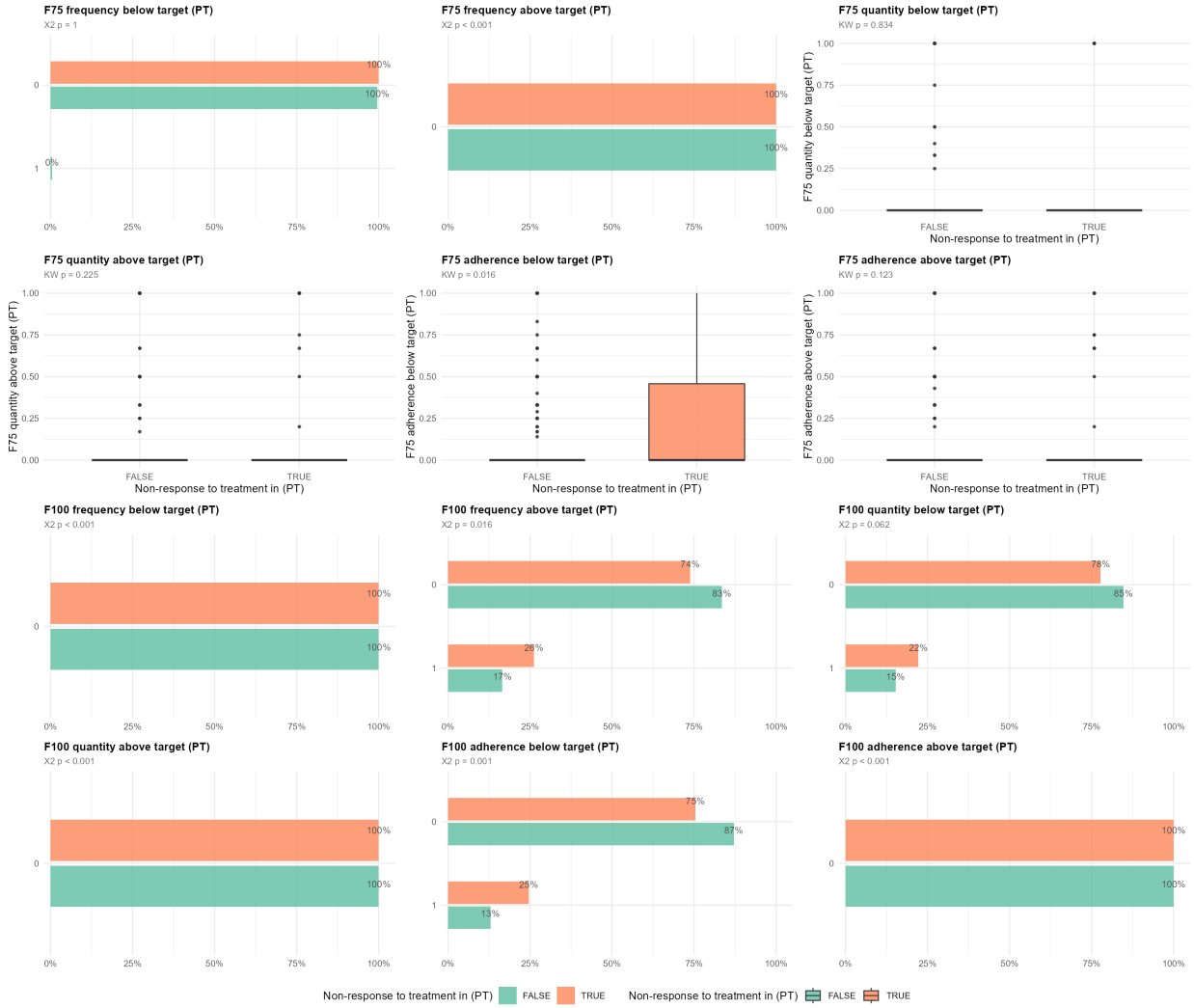
Clinical Admission and Severity



Nutritional Treatment

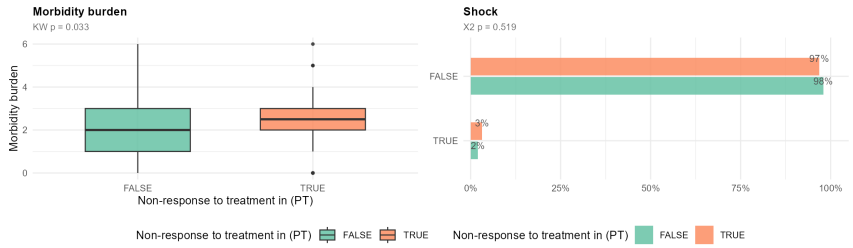


Feeding Compliance

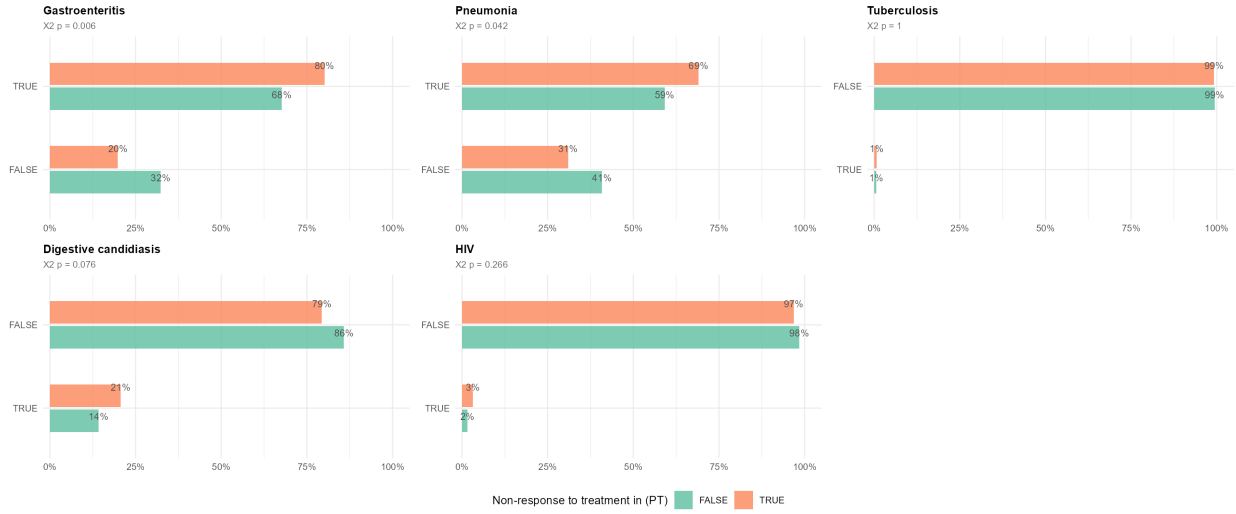


Morbidity and Comorbidities

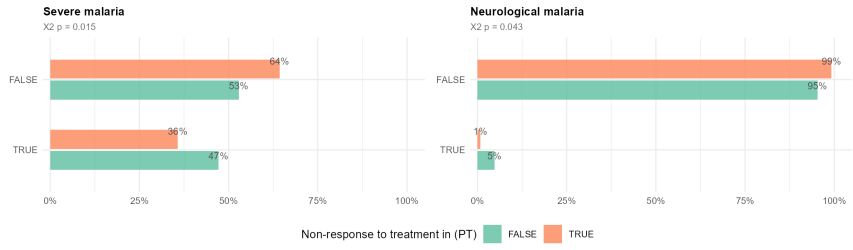
General



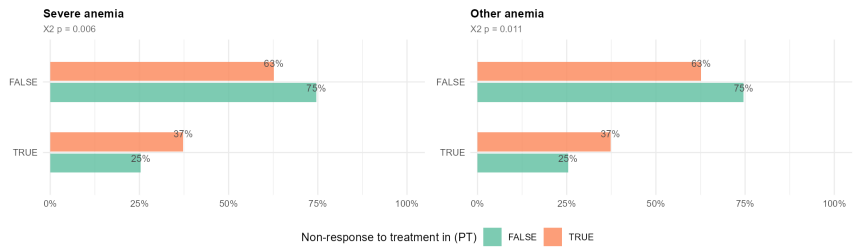
Infections and Parasitic



Malaria



Anaemia



Insert comments here.

Table 2: Results of the univariate logistic regression analysis between explanatory variables and outcome.

Predictor	Event N	OR	95% CI	p-value
Gastroenteritis				**0.004**
FALSE	25	—	—	
TRUE	101	1.93	1.23, 3.14	
Severe anemia				**0.008**
FALSE	79	—	—	
TRUE	47	1.75	1.16, 2.61	
Severe malaria				**0.018**
FALSE	81	—	—	
TRUE	45	0.62	0.42, 0.92	
Pneumonia				**0.035**
FALSE	39	—	—	
TRUE	87	1.54	1.03, 2.34	
F75 adherence below target (PT)				0.051
0	92	—	—	
]0;100]	34	1.57	1.00, 2.42	
Food intolerance proportion (P1)	126	1.46	0.99, 2.15	0.058
Digestive candidiasis				0.076
FALSE	100	—	—	
TRUE	26	1.57	0.95, 2.53	
Family meals proportion (P1)	126	6.12	0.49, 69.3	0.15
F75 adherence above target (PT)				0.2
0	111	—	—	
]0;100]	15	1.54	0.81, 2.78	
HIV				0.3
FALSE	122	—	—	
TRUE	4	2.05	0.55, 6.23	
Shock				0.5
FALSE	122	—	—	
TRUE	4	1.57	0.44, 4.51	
Type admission				0.6
New admission	110	—	—	
Readmission	16	0.87	0.48, 1.50	
Tuberculosis				0.8
FALSE	125	—	—	

Predictor	Event N	OR	95% CI	p-value
TRUE	1	1.26	0.06, 8.61	
Breastfeeding status				>0.9
No	5	—	—	
Yes	106	1.04	0.42, 3.13	

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Insert comments here.

5. Multivariate logistic regression analyses

Table 3 : Results of the generalize logistic regression analysis at Phase T.

Predictor	N	Event N	OR[1]	95% CI	p-value	q-value[2]	VIF
Gastroenteritis	760		*		0.014	**0.053**	1.0
FALSE		25		—			
TRUE		101	1.77	1.12, 2.89			
Severe anemia	760		*		0.026	**0.053**	1.0
FALSE		79		—			
TRUE		47	1.60	1.06, 2.40			
Pneumonia	760				0.068	**0.074**	1.0
FALSE		39		—			
TRUE		87	1.46	0.97, 2.24			
F75 adherence below target (PT)	760				0.074	**0.074**	1.0
0		92		—			
]0;100]		34	1.52	0.96, 2.35			

[1]*p<0.05; **p<0.01; ***p<0.001

[2]False discovery rate correction for multiple testing

Abbreviations: CI = Confidence Interval, OR = Odds Ratio, VIF = Variance Inflation Factor

Insert comments here.